Data Science and Artificial Intelligence (DA) :

1. **Probability and Statistics**:
    - Counting (permutations and combinations)
    - Probability axioms
    - Sample space, events
    - Independent events, mutually exclusive events
    - Marginal, conditional, and joint probability
    - Bayes' Theorem
    - Conditional expectation and variance
    - Measures of central tendency: mean, median, mode
    - Measures of dispersion: standard deviation, correlation, covariance
    - Random variables
    - Discrete random variables and probability mass functions
    - Continuous random variables and probability distribution functions (uniform, exponential, Poisson, normal, standard normal, t-distribution, chi-squared distributions)
    - Cumulative distribution function
    - Conditional probability density function
    - Central limit theorem
    - Confidence intervals
    - Hypothesis testing: z-test, t-test, chi-squared test

2. **Linear Algebra**:
    - Vector space, subspaces
    - Linear dependence and independence of vectors
    - Matrices: projection matrix, orthogonal matrix, idempotent matrix, partition matrix
    - Quadratic forms
    - Systems of linear equations and solutions
    - Gaussian elimination
    - Eigenvalues and eigenvectors
    - Determinant, rank, nullity
    - Projections
    - LU decomposition
    - Singular value decomposition

3. **Calculus and Optimization**:
    - Functions of a single variable
    - Limit, continuity, and differentiability
    - Taylor series
    - Maxima and minima
    - Optimization involving a single variable

4. **Programming, Data Structures, and Algorithms**:
    - Programming in Python
    - Basic data structures: stacks, queues, linked lists, trees, hash tables
    - Search algorithms: linear search, binary search
    - Basic sorting algorithms: selection sort, bubble sort, insertion sort
    - Divide and conquer: mergesort, quicksort
    - Introduction to graph theory
    - Basic graph algorithms: traversals, shortest path

5. **Database Management and Warehousing**:
    - ER-model
    - Relational model: relational algebra, tuple calculus, SQL
    - Integrity constraints
    - Normal forms
    - File organization
    - Indexing
    - Data types
    - Data transformation: normalization, discretization, sampling, compression
    - Data warehouse modeling: schema for multidimensional data models, concept hierarchies, measures

6. **Artificial Intelligence (AI)**:
    - Search algorithms: informed, uninformed, adversarial
    - Logic: propositional, predicate
    - Reasoning under uncertainty: conditional independence representation, exact inference through variable elimination, approximate inference through sampling

7. **Machine Learning**:

- Supervised Learning:
  - Regression and classification problems
  - Linear regression (simple and multiple)
  - Ridge regression
  - Logistic regression
  - k-Nearest Neighbors
  - Naive Bayes classifier
  - Linear discriminant analysis
  - Support vector machine
  - Decision trees
  - Bias-variance trade-off
  - Cross-validation methods (leave-one-out, k-folds)
  - Multi-layer perceptron, feed-forward neural network

- Unsupervised Learning:
  - Clustering algorithms: k-means, k-medoid
  - Hierarchical clustering (top-down, bottom-up)
  - Dimensionality reduction: principal component analysis